Pilanilabs Career Platform — End-to-End Product Spec

Oct 27, 2026 · @Ajay

1. Executive summary

The new product joins a full job-search workspace (already built as the Pilanilabs Job Hunting Tool) with an outreach engine like HAPPPY's: paste a job link, and the system finds the hiring team, drafts personalised messages and runs the email and LinkedIn follow-up for the candidate.

The problem. Candidates apply to dozens of jobs and hear nothing. Most roles are filled through referrals and direct contact with the hiring manager, but finding that person, writing a good message and following up on time is slow, so most candidates never do it.

The product in one sentence. One place where a job seeker discovers fresh jobs, gets a resume tailored to each one, checks it against ATS filters, and then reaches the people who can hire them, with every step tracked.

Who it is for.

- Primary: job seekers in India, from freshers to about 15 years of experience, in tech, product, design, marketing and operations.
- Secondary: global candidates looking for remote roles.
- Later: career coaches, colleges and placement cells managing many candidates.

Why it can win. HAPPPY does outreach only: the candidate must bring the job and the resume. This product owns the whole journey, so the outreach is fed by ranked jobs, a tailored resume and a tracked pipeline, and the reply data improves the matching.

Three decisions that shape everything (detail in section 8):

1. How far outreach is automated on LinkedIn, whose terms prohibit bots and automation tools.
2. How email is sent: from the candidate's own Gmail, or from a managed sending domain.
3. Which licensed data provider supplies hiring-manager contacts, since scraping is not an option.

2. What is already built

The Pilanilabs Job Hunting Tool is deployed on Vercel with a Supabase backend, has 79 unit tests and 8 end-to-end runs (desktop and 375 px mobile), and covers the whole search journey except automated outreach.

Module	What it does today
Sign-in	Email link plus a typed 6-digit code (works across browsers and devices), and Google sign-in. Row Level Security on all user tables.
Resume intake	PDF or DOCX up to 5 MB, checked by file signature, stored in a private bucket behind signed URLs. Parsed by an LLM into a structured profile the user reviews and edits. Offline fallback parser if the AI fails.
Preferences	Target roles, locations, work mode, job type, minimum salary, notice period, include and exclude keywords.
Job sources	Nine connectors: JSearch (Google for Jobs), Adzuna, Remotive, RemoteOK, Arbeitnow, Jobicy, and public Greenhouse, Lever and Ashby boards. Failures are isolated per source. Sources are credited and link back.
Dedupe	One record per job from a hash of title, company and city; the most complete record wins and every source seen is kept.
Matching	Stage 1 (no AI): filters plus a 0-100 score from skills, title, experience and location, keeping the top 200. Stage 2 (AI): re-scores the best 50 with a reason, matching and missing skills. Cached by profile hash.
Feed	Filters, sort, search, infinite scroll, skeleton loaders, save to tracker.
Job detail	Full description, match breakdown, ATS check, referral helper, cover letter, apply link in a new tab with an "Did you apply?" confirmation.

Module	What it does today
Tailored resume	Rewrites the summary, reorders skills and bullets for one job. Never adds facts: output is checked in code against the original and violations are removed. Side-by-side compare, inline edit, PDF and DOCX export that real parsers can read.
ATS checker	Parse checks, format checks and keyword coverage against a job description. Deterministic, no AI.
Hidden jobs	Openings found only on company career pages; users can add any Greenhouse, Lever or Ashby board by URL.
Company pages	Open roles, curated facts, outbound research links, and reviews written by users.
Tracker	Kanban: Saved, Applied, Interviewing, Offer, Rejected, with drag and drop, notes and follow-up dates.
Alerts	Saved searches with names; daily or weekly digests by email (Resend) and WhatsApp (Twilio).
Referral helper	Drafts a LinkedIn connection note (300 characters) and a follow-up. Search links are provided; the user finds the person and sends it. Nothing is automated.
Account	Multiple resumes, delete account and all data, download all data as JSON, privacy page.

Stack. Next.js 16 (App Router) and TypeScript, Tailwind CSS 4, Supabase (Postgres, Auth, Storage), Prisma 7, Anthropic Claude through one provider layer that can also use free OpenRouter models, Zod validation on every input and AI output, Vercel hosting and cron.

Known limits today.

- Not yet exercised with live credentials: JSearch, Adzuna, Resend, Twilio and the OpenRouter models. They are tested against fakes only.
- Free AI models have small shared daily quotas and lower quality than Claude.
- Vercel Hobby runs crons once a day; Gmail SMTP is used for sign-in emails.
- There is no contact discovery, no email sending on behalf of the user, no reply tracking and no team or coach accounts.

3. HAPPPY AI Agent (Uplers) analysis

HAPPPY is a referral and outreach agent: the candidate pastes a job link and the agent contacts people at the company on the candidate's behalf, so that a recruiter replies. Everything below is from the public landing page (HAPPPY on Uplers, read 27 Oct 2026) and the product owner's brief; the page's performance claims are the vendor's own and unverified.

How it works, as the page describes it

1. Configure the agent in about 60 seconds by connecting a Gmail account and, optionally, LinkedIn.
2. Run the agent on a job: paste any job URL from LinkedIn, Indeed or a company career site, or pick a job inside the platform.
3. The agent finds relevant referral contacts at the company (team leads, hiring managers, recruiters) and reaches out.
4. The candidate gets an insider referral and, the page claims, an interview.

Channels

Channel	What the page says	Product owner's brief
Gmail	Any Gmail account can be connected and disconnected at any time. Sends the resume directly to the recruiter's official inbox and asks team leads and hiring managers for referrals.	Finds the hiring manager's contact details and emails them; emails can be scheduled and followed up.
LinkedIn	Optional, no Premium needed. Auto-sends connection requests to people at target companies and sends the resume by InMail. The page claims up to 2x replies when LinkedIn is added to Gmail.	Sends a connection request first; only after it is accepted does a message go out.

Claims and commercial model (vendor's own)

- Interviews "in as little as 4 days", "2x interviews", "trusted by 6450+ candidates", live counter of interviews in the last 7 days, a wall of employers where interviews were reported.
- Pay when it works: "Try until you hear yes. Upgrade only when someone replies", cancel anytime, with monthly and quarterly plans.

What the public page does not show, and we must decide ourselves

- Where contact data comes from, and how emails are verified.
- Follow-up cadence, send limits per day, and how replies are detected and handled.
- How LinkedIn is driven (official API, the candidate's browser session, or a third-party tool) and how accounts are protected from restrictions.
- How consent, opt-outs and data deletion are handled for the people being contacted.

Where this product differs. HAPPPY starts at a job the candidate already chose. This product also finds and ranks the jobs, tailors the resume to each, checks ATS fit, and tracks the whole pipeline, then runs outreach from that context.

4. Product requirements

The new product has two halves: the existing search workspace (carried over and hardened) and the new outreach engine. MVP means the first paid release; v2 follows once reply data exists; Later is unscheduled.

A. Search workspace (carry over)

ID	Requirement	Priority
A1	Resume upload, AI parsing, editable profile and preferences	MVP
A2	Multi-source job feed with dedupe, two-stage matching and filters	MVP
A3	Hidden jobs from company career pages, user-added boards	MVP
A4	Per-job tailored resume and cover letter with no-fabrication guard, PDF and DOCX export	MVP
A5	ATS score checker for any resume and job description	MVP
A6	Application tracker, now also showing outreach status per job	MVP
A7	Saved searches and email or WhatsApp alerts	MVP
A8	Company pages with open roles, facts and reviews	v2

B. Outreach engine (new)

ID	Requirement	Priority
B1	Add a job by pasting a link (LinkedIn, Indeed, career sites) or by choosing one from the feed; extract title, company, location and description	MVP
B2	Resolve the company and identify target people: hiring manager, team lead, recruiter, plus alumni or ex-colleagues as warm paths	MVP
B3	Contact discovery and email verification through licensed data providers, with confidence score and source shown	MVP
B4	Candidate-approved message drafts: one email and one LinkedIn note per target, personalised from the resume, the tailored resume and the job	MVP
B5	Email sequence: initial email plus up to two follow-ups, scheduled in the candidate's time zone and the recipient's working hours, with daily send caps	MVP
B6	Send from the candidate's own connected mailbox (Gmail first, Outlook later) with the tailored resume attached	MVP
B7	Reply detection: stop the sequence on any reply, bounce or unsubscribe; classify the reply (interested, referral, not now, no) and notify the candidate	MVP
B8	LinkedIn: connection request first, message only after acceptance (mode depends on decision 1 in section 8)	v2
B9	Outreach inbox: one thread view per job with email and LinkedIn history, next scheduled step, edit or cancel any step	MVP
B10	Suppression list and do-not-contact handling; one-click opt-out line in every email	MVP
B11	Autopilot levels: review every message, approve the first message only, or fully automatic within limits	v2
B12	Interview prep triggered by a positive reply (company brief, likely questions)	v2

ID	Requirement	Priority
B13	Outcome tracking: reply rate, referral rate, interview rate per template and per company type	MVP
B14	Coach and college accounts managing many candidates, with a shared sending policy	Later

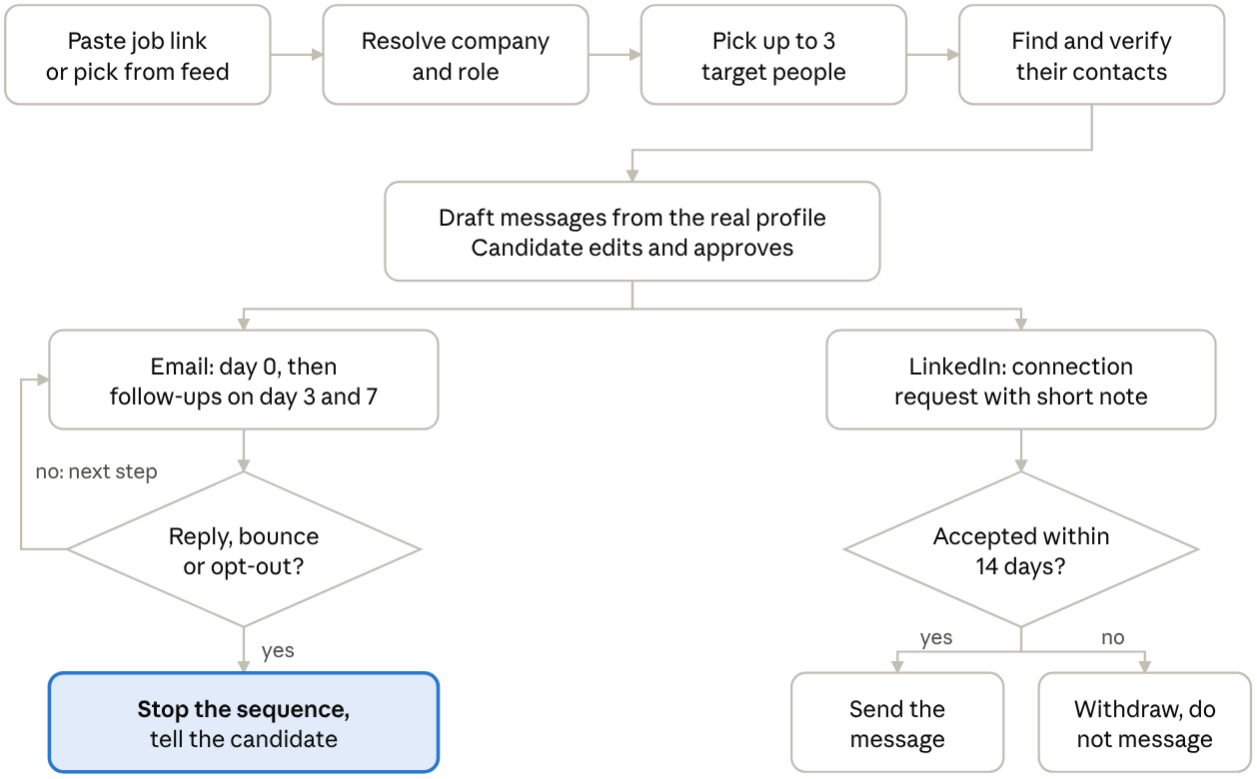
C. Platform

ID	Requirement	Priority
C1	Billing: free tier for the workspace; outreach on credits or pay-per-reply (section 10)	MVP
C2	Admin console: provider costs, send health, complaint rates, abuse controls	MVP
C3	Audit log of every message sent and every consent given	MVP
C4	Mobile-first web app and installable PWA; native apps later	v2

5. Outreach automation engine

The engine turns one pasted job link into a scheduled, tracked, multi-channel outreach plan that the candidate approves, and stops itself the moment anyone replies.

One approval starts two lanes, and any reply stops the sequence



Whether LinkedIn steps are assisted or automated is decision 1 in section 8

outreach pipeline · 10 steps, 2 decisions

Every contact passes the same first six steps; after approval the email lane and the LinkedIn lane run side by side, and any reply, bounce or opt-out ends the email sequence for that person.

Pipeline, step by step

1. **Intake.** The candidate pastes a job URL. The system fetches the posting (or uses the stored job if it came from the feed), extracts title, company, location and requirements, and links it to the candidate's tailored resume, creating one if missing.
2. **Company resolution.** Match the employer to a company record (domain, size, industry, recent hiring) so contacts can be looked up and the same company is never contacted twice in 30 days by one candidate.
3. **Target selection.** Rank people by likely usefulness: the hiring manager for the team, then team leads, then recruiters and talent acquisition, then warm paths (same college, former colleague). Default is 3 targets per job.
4. **Contact discovery and verification.** Query licensed data providers for each person's work email and LinkedIn profile, verify deliverability, and store the source and confidence. Contacts below the confidence threshold are shown but not sent without an explicit candidate decision.

5. **Drafting.** An LLM writes one email and one LinkedIn note per target from the candidate's real profile, the tailored resume and the job. Rules: use only facts from the resume, never claim a mutual connection that was not found, no flattery, under 120 words, a clear low-pressure ask.
6. **Approval.** The candidate sees every draft, edits it, and approves it (autopilot levels come in v2). Nothing is sent without an approval record.
7. **Scheduling and sending.** Emails go from the candidate's connected mailbox inside the recipient's local working hours, spaced out, within daily caps. Sequence: initial email on day 0, follow-up on day 3, final note on day 7.
8. **LinkedIn step (see decision 1).** Send a connection request with a short note first. Only when the request is accepted, send the message; if there is no acceptance after 14 days, withdraw the request and do not message.
9. **Reply handling.** Poll the mailbox for replies and bounces. Any reply, bounce or opt-out pauses all further steps to that person. The reply is classified and the candidate is notified at once with a suggested response.
10. **Learning.** Every outcome (sent, opened if tracked, replied, referral, interview) feeds the dashboard and improves target ranking and templates.

Safety rules built into the engine

- Hard daily caps per mailbox with a warm-up ramp for new mailboxes; per-recipient and per-company frequency limits.
- Every email carries an honest sender identity and a simple opt-out line; opted-out addresses go on a global suppression list that outlives account deletion.
- Bounce rate or spam-complaint rate above a threshold automatically pauses the candidate's sending.
- No scraping of LinkedIn or any site that forbids it; contacts come only from licensed providers or from information the candidate supplies.
- A full audit log: who approved what, what was sent, when, from which mailbox.

Per-target states. Discovered, verified, drafted, approved, scheduled, sent, then one of replied, bounced, opted out, completed or cancelled. LinkedIn adds connect pending, connected and messaged.

6. User journeys and screens

A new candidate should reach their first approved outreach in under 10 minutes, and every later job in under 2.

Journey 1: first session

1. Land on the home page, sign in by email code or Google.

2. Upload a resume; review the parsed profile; set preferences.
3. See a ranked feed within about a minute.
4. Connect a mailbox (Gmail) and, if the chosen mode allows it, LinkedIn. Choose a sending name and a daily cap; the product explains what will and will not be sent.

Journey 2: outreach from a pasted link

1. Paste a job link on the Outreach page (or press "Reach out" on any feed job).
2. Review the extracted job; the system proposes a tailored resume and three target people with sources and confidence.
3. Review drafts, edit, approve. Preview the schedule: which email goes when, in whose time zone.
4. The job moves to "Outreach active" in the tracker. The candidate can pause or cancel any step.

Journey 3: a reply arrives

1. Notification by email, WhatsApp or push: who replied, what they said, the reply type.
2. The sequence to that person stops. The candidate answers with an AI-suggested reply or opens the thread in Gmail.
3. The job moves to Interviewing when the candidate confirms; interview prep is offered.

Journey 4: nothing happens

After the sequence ends with no reply, the candidate is shown the result and offered the next three targets, or a different job, with a note on what to change.

Screen list

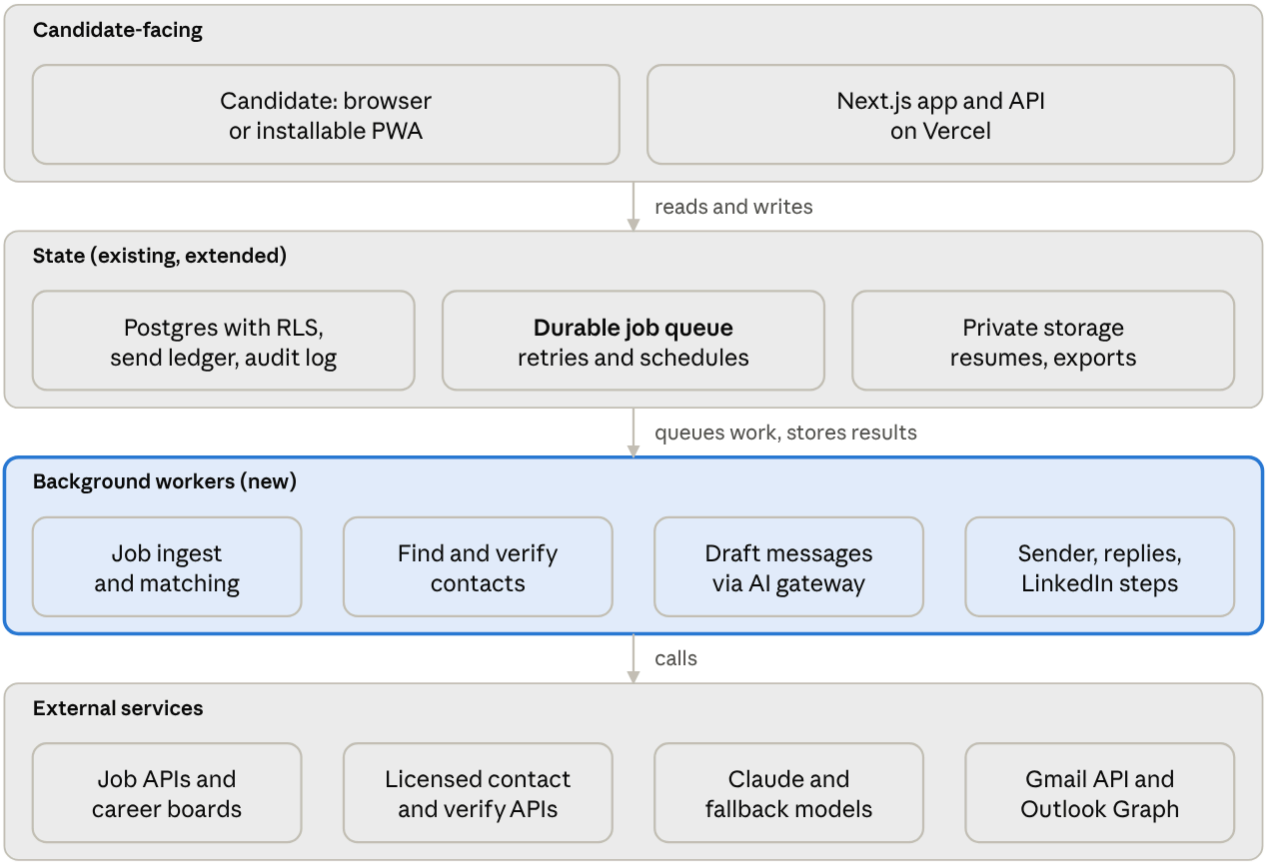
Screen	Purpose
Home, sign-in	Value proposition, email code and Google sign-in
Onboarding	Upload, review profile, preferences, connect mailbox
Feed and job detail	Ranked jobs, match breakdown, ATS check, tailor, reach out
Tailor	Compare, edit, export resume; cover letter
Outreach new	Paste link, extracted job, targets, drafts, approval, schedule preview
Outreach inbox	One thread per job and person; status, next step, replies

Screen	Purpose
Tracker	Kanban of applications with outreach status
Hidden jobs, Companies, Saved searches	Discovery tools carried over
Insights	Reply, referral and interview rates; what is working
Settings	Resumes, preferences, connected accounts, send limits, data export and deletion
Admin (internal)	Provider spend, send health, complaints, suppression list, abuse review

7. Architecture, data and integrations

The workspace stays a Next.js app on Supabase; the outreach engine adds a durable job queue and background workers, because sending, polling and follow-ups cannot run inside short web requests.

The outreach engine adds a queue and workers behind the existing app



LinkedIn has no public API for this, so LinkedIn steps stay assisted unless decision 1 changes

system architecture · 4 layers

The web app only reads, writes and queues; every slow or scheduled action runs in a worker that can retry safely.

Components

- **Web app (Next.js on Vercel):** screens, API routes, authentication, approvals.
- **Queue and workers:** a Postgres-backed queue (for example Graphile Worker or pg-boss) or a managed one (Inngest, Trigger.dev, QStash). Workers: job ingest and matching, contact discovery, drafting, scheduled sender, reply poller, LinkedIn step runner.
- **Postgres (Supabase):** all state, with Row Level Security; a send ledger with unique keys so a message can never be sent twice.
- **Object storage:** resumes and exports in private buckets.
- **AI gateway:** one layer that calls Claude or fallback models, validates every output with schemas, logs tokens and cost per user.
- **Mailbox connectors:** Gmail API (send, read replies) with OAuth, then Microsoft Graph for Outlook.

- **Contact data connectors:** licensed providers behind one interface, so a provider can be swapped.
- **Observability:** structured logs, send-health dashboard, alerts on bounce and complaint spikes.

Pasting a job link. Known career systems (Greenhouse, Lever, Ashby, Workable) are read through their public APIs; other pages are read from their structured JobPosting data or by fetching and extracting the text. LinkedIn and Indeed pages are not scraped: for those links the product looks the job up through a licensed job-search API or asks the candidate to paste the description.

Data model additions (on top of the current User, Resume, Preferences, Job, Match, TailoredResume, Application, SavedSearch tables)

Table	Holds
ConnectedAccount	Provider, account email, encrypted refresh token, granted scopes, status, daily send cap, warm-up stage
OutreachCampaign	One per candidate and job: tailored resume used, mode, status, created and finished times
OutreachTarget	Person name and title, company, source of the data, confidence, verified email and result, LinkedIn URL, state
OutreachMessage	Channel, sequence step, subject, body, approval record, scheduled and sent times, provider message and thread ids, state
Reply	Linked message, received time, classification, suggested response
SuppressionEntry	Hash of an email address that must never be contacted again, with reason; kept after account deletion
AuditEvent	Who approved, sent, paused or deleted what, and when
CreditLedger	Credits bought, spent per target or per verified contact, refunds, pay-per-reply events
ProviderUsage	Calls and cost per data provider per candidate, for margin control

Security. Mailbox tokens are encrypted with envelope encryption; scopes are the minimum needed; tokens are revoked on disconnect or account deletion; secrets never reach the browser; every outbound message needs a stored approval; admin access is logged.

8. Compliance, platform policy and the decisions they force

The outreach half is where the product can be harmed, so three choices must be made before building it: how LinkedIn is handled, how email is sent, and where contacts come from. This section is a product risk summary, not legal advice; have counsel review it before launch.

Decision 1: LinkedIn automation

LinkedIn's User Agreement prohibits bots and automated tools that send connection requests or messages, and LinkedIn restricts accounts that use them. It also has no public API for ordinary developers to send connection requests or messages. HAPPPY's page describes exactly this automation, so matching it means taking on that risk.

Option	How it works	Risk	Recommendation
A. Assisted (compliant)	The product finds the person, writes the connection note, tracks the 14-day wait, prepares the message for after acceptance, reminds the candidate and opens the right profile. The candidate clicks Send in LinkedIn.	Low. Less magic, more candidate effort.	Ship in MVP
B. Official partner access	Use LinkedIn's partner programs if the company qualifies.	Low, but access is limited and slow to obtain.	Explore in parallel

Option	How it works	Risk	Recommendation
C. Session automation	A browser extension or cloud browser acts inside the candidate's logged-in account.	High: account restrictions for candidates, legal and reputational exposure for the company.	Do not ship by default; only with explicit informed opt-in, if the company accepts the risk

Decision 2: how email is sent

Option	Notes
A. Candidate's own Gmail through OAuth	Best deliverability and replies land in the candidate's inbox. Sending needs a sensitive Google scope; reading replies needs a restricted scope, which requires Google's OAuth verification and an annual third-party security assessment. Plan for weeks of lead time.
B. Product-owned sending domain (Amazon SES, Postmark) with the candidate's name and a reply-to address	Faster to launch and no Google review, but shared-domain reputation risk, weaker trust, and recipients may see it as impersonation. Needs a strict policy.
C. Candidate's SMTP or app password	Fragile and unsafe to store; avoid.

Recommendation: A for production, B only as a stop-gap clearly labelled "sent on behalf of".

Decision 3: where contacts come from

Use licensed contact-data and email-verification providers behind one interface; do not scrape LinkedIn or other sites that forbid it. Providers built on scraped data carry legal and continuity risk, so ask each how its data is sourced. Coverage in India is weaker than in the US, which affects the promise made in marketing. Budget per verified contact and cap it per candidate.

Regulatory and policy checklist

Area	Risk	Control in the design
CAN-SPAM (US)	Commercial email needs honest sender identity, no deceptive subject, a physical address and a working opt-out honoured within 10 business days	Candidate's real name and address of record, plain subjects, opt-out line, suppression list
GDPR and ePrivacy (EU, UK)	Contacting EU individuals needs a lawful basis; some countries restrict unsolicited email even to work addresses	Region filter, excluded by default until a lawful-basis review is done
India DPDP Act 2023	Personal data of contacts and candidates must be processed lawfully, with notice, security and deletion on request	Purpose limitation, retention limits, deletion and access requests, India-hosted data option
Google API policy	Restricted-scope data may only be used to provide the feature the user sees; no ad use or human reading except narrow cases	Minimal scopes, no model training on mail content, security assessment
Candidate harm	Fabricated claims, spamming a company, damaging the candidate's reputation	No-fabrication guard, one touch per company per 30 days, caps, approval record
Data deletion	Candidate leaves but recipients must stay opted out	Delete personal data, keep only hashed suppression entries

9. AI design and guardrails

AI writes, scores and classifies; code decides who is contacted, what is sent and when. The model never picks a recipient and never sends anything.

AI tasks

Task	Model class	Output	Guardrail
Parse resume	Quality	Structured profile (schema)	Contact fields must appear in the source text; offline fallback parser

Task	Model class	Output	Guardrail
Extract job from a page	Fast	Title, company, location, requirements	Schema check; page text treated as untrusted data
Score job fit	Fast	0-100, reason, matching and missing skills	Stage-1 code filters run first; scores cached per profile
Tailor resume, cover letter	Quality	Resume in the same schema, change list, skill gaps	Verifier rejects any employer, title, date, degree, certificate, skill or number not in the original; auto-repair
Rank outreach targets	Rules first, fast model to break ties	Ordered list with reason	Only people returned by licensed providers
Draft email and LinkedIn note	Quality	Subject, body, note under 300 characters	Only facts from the profile; no claimed mutual connection unless found; length and tone limits; approval required
Classify a reply	Fast	Interested, referral, not now, no, out of office, unsubscribe	Unsubscribe or hostile replies always stop the sequence by rule, whatever the model says
Suggest a response, interview prep	Quality	Draft text	Shown to the candidate, never auto-sent

Safeguards that apply everywhere

- Every output is validated against a schema; invalid output gets one corrective retry, then a clear error or a deterministic fallback.
- Prompt-injection defence: job pages, contact profiles and email replies are data, never instructions; model output cannot trigger a send, a recipient change or a tool call.
- Message verifier: names, employers, schools and numbers in a draft must exist in the candidate's profile; the same repair approach used for resumes.

- Send requires an approval record with the exact text that was approved; edits after approval require re-approval.
- Personal data sent to a model is limited to what the task needs; no model training on user data; per-user rolling daily limits on every AI feature.

Evaluation before launch

- A golden set of about 200 resume and job pairs checked for fabrication (target: zero), schema validity and tailoring quality by human reviewers.
- Message quality scored by reviewers on a rubric (specific, honest, short, polite); then reply rate per template measured live.
- Regression suite run on every prompt or model change.

Cost control. Cache every AI output, use the fast model for bulk work, cap per-user daily AI calls, track AI and data-provider cost per campaign, and set a hard per-candidate cost ceiling. Measure the real cost of one outreach campaign in the pilot before fixing prices (section 10).

10. Business model and metrics

Give the search workspace away to build the user base, and charge for outreach, where the data and sending costs are. Prices are left open until a pilot measures real cost per campaign.

Pricing options

Model	How it works	Pros	Cons
Free workspace plus outreach credits	One credit per verified target contacted; packs and a monthly plan	Simple, ties price to the real cost	Candidate pays before knowing it works
Pay on reply (HAPPPY's approach: upgrade only when someone replies)	Free to run; pay when a reply arrives	Low friction, strong trust	Company carries all data and sending cost; invites gaming and disputes over what counts as a reply
Subscription	Monthly or quarterly plan with a send allowance	Predictable revenue	Hard to justify before first results

Model	How it works	Pros	Cons
Hybrid (recommended to test)	Free workspace and a small free outreach trial; credits after that; optional "pay on reply" bundle	Balances trust and cost	More to build and explain

Margin per outreach campaign

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\text{margin} = \text{price} - (\text{AI cost} + \text{contact data} + \text{email verification} + \text{infrastructure} + \text{support})
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Metrics to track from day one

Metric	Why it matters
Activation: resume uploaded to first approved outreach	Shows whether onboarding works
Reply rate, referral rate, interview rate per campaign	The product's promise; compare against the vendor claims above
Time to first reply	Candidates judge the product on days, not weeks
Bounce rate and spam-complaint rate per mailbox	Protects deliverability and Google account standing; pause thresholds
Verified-contact hit rate by country	Tells whether the India promise is honest
AI plus data cost per campaign, and gross margin	Decides pricing
Candidate retention and referrals	Growth through outcomes

Open commercial questions: price points for India versus global users, whether colleges and coaches are a first channel, and whether pay-on-reply is worth its disputes.

11. Roadmap, team and next steps

A sensible path reaches a paid, compliant email-outreach pilot in about 14 weeks, because the search workspace already exists; LinkedIn automation and coach accounts come after reply data proves the value. Durations are planning estimates, not commitments.

Phase	Weeks	Scope	Exit gate
0. Decide and de-risk	0-2	Decisions 1-3 (section 8), counsel review, evaluate two or three contact-data providers on India coverage and cost, start Google OAuth verification, choose queue technology	Written decisions; provider shortlisted; Google review submitted
1. Foundation	2-8	Queue and workers, connected mailbox, job-link intake, target selection, contact discovery and verification, drafting, approval, scheduled email sequence, suppression list, audit log. Internal testers send from their own mailboxes.	20 internal campaigns sent with zero duplicate sends and bounce rate under the pause threshold
2. Pilot	8-14	Reply detection and classification, outreach inbox, tracker integration, credits and billing, admin console, send-health alerts. Invite about 50 candidates.	Measured reply and referral rates, cost per campaign, complaint rate; go or no-go
3. Grow	14-22	LinkedIn assisted mode, autopilot levels, insights, interview prep, Outlook mailbox, region support	Retention and referral growth; margin positive per campaign
4. Scale	22+	Coach and college accounts, LinkedIn partner access if available, native mobile apps, more countries	Decided from pilot data

Team (planning assumption). Two full-stack engineers, one backend and data engineer for queue, connectors and deliverability, a part-time designer, a product owner, and external counsel for the compliance review.

Open decisions

- [ ] Decision 1: LinkedIn mode for MVP (recommended: assisted).
- [ ] Decision 2: email path (recommended: candidate's Gmail through OAuth; managed domain only as a stop-gap).

- [ ] Decision 3: contact-data provider and the per-candidate cost cap.
- [ ] Pricing model for the pilot.
- [ ] Launch regions; whether the EU and UK are excluded at first.
- [ ] Whether to keep the current codebase as the base (recommended: yes, add workers and the outreach tables) or rebuild.

Next steps

1. Hold a decision meeting on the three decisions above, with counsel.
2. Run a two-week provider bake-off using 50 real job links and measure verified-contact hit rate in India.
3. Submit the Google OAuth verification for the mailbox scopes as early as possible, since it has the longest lead time.
4. Build Phase 1 on the existing Pilanilabs codebase.

12. API keys and environment variables (names only)

This list names every credential the system needs and where to get it. It contains no values: never put real keys in a prompt, a document or a repository. Real values live only in the hosting provider's secret store and in a local, git-ignored env file.

Variable	Purpose	Where it comes from	Secret?
NEXT_PUBLIC_APP_URL	Public base URL for links and redirects	Your domain	No
DATABASE_URL	Pooled Postgres connection for the app	Supabase, Project Settings, Database	Yes
DIRECT_URL	Direct Postgres connection for migrations	Supabase, Project Settings, Database	Yes
NEXT_PUBLIC_SUPABASE_URL	Supabase project URL	Supabase, Project Settings, API	No

Variable	Purpose	Where it comes from	Secret?
NEXT_PUBLIC_SUPABASE_ANON_KEY	Public client key (publishable key)	Supabase, API Keys	No (public by design)
SUPABASE_SERVICE_ROLE_KEY	Server-only key for private storage and account deletion	Supabase, API Keys	Yes
SUPABASE_STORAGE_BUCKET	Private bucket name for resumes and exports	Chosen by you	No
CRON_SECRET	Protects scheduled endpoints	Generate a long random string	Yes
ANTHROPIC_API_KEY	Claude for parsing, tailoring, drafting, scoring	Anthropic Console	Yes
AI_PROVIDER, OPENROUTER_API_KEY, OPENROUTER_MODEL_QUALITY, OPENROUTER_MODEL_FAST	Optional free-model provider and its model chains	OpenRouter	Key: yes
AI_MATCH_SCORING	Turn AI feed scoring on or off	Chosen by you	No
RAPIDAPI_KEY	JSearch job search (Google for Jobs)	RapidAPI	Yes
ADZUNA_APP_ID, ADZUNA_APP_KEY	Adzuna job search	Adzuna developer portal	Key: yes
JOB_SOURCES_CONTACT	Contact URL sent in the User-Agent of public API calls	Your site	No

Variable	Purpose	Where it comes from	Secret?
RESEND_API_KEY, RESEND_FROM	Alert emails	Resend (verified sender domain)	Key: yes
TWILIO_ACCOUNT_SID, TWILIO_AUTH_TOKEN, TWILIO_WHATSAPP_FROM, TWILIO_WHATSAPP_CONTENT_SID	WhatsApp alerts	Twilio	Token: yes
DEV_AUTH_BYPASS	Local-only sign-in shortcut; must never be set in production	Chosen by you	No
GOOGLE_OAUTH_CLIENT_ID, GOOGLE_OAUTH_CLIENT_SECRET (new)	Connecting a candidate's Gmail for sending and reply detection; also Google sign-in	Google Cloud Console	Secret: yes
MICROSOFT_OAUTH_CLIENT_ID, MICROSOFT_OAUTH_CLIENT_SECRET (new, v2)	Outlook mailbox connection	Microsoft Entra	Secret: yes
CONTACT_DATA_API_KEY (new)	Licensed contact-discovery provider	The provider chosen in the bake-off	Yes
EMAIL_VERIFY_API_KEY (new)	Email deliverability verification	Verification provider	Yes
TOKEN_ENCRYPTION_KEY (new)	Encrypts stored mailbox tokens (or use a managed key service)	Generate 32 random bytes, or cloud KMS	Yes

Variable	Purpose	Where it comes from	Secret?
QUEUE_URL or QUEUE_TOKEN (new)	Durable job queue, if a managed one is used	Queue provider	Yes
SENTRY_DSN (new, optional)	Error reporting	Sentry	No

Rules for anyone building from this document

- Use placeholders such as `YOUR_KEY_HERE` in any prompt, example or commit; never paste a real value.
- Add a startup check that fails loudly in production when a required secret is missing, and never log secret values.
- Keep secrets out of client bundles: only variables starting with NEXT_PUBLIC_ are exposed to the browser.
- Rotate any key that has ever appeared in a chat, screenshot or ticket.

13. Database schema (PostgreSQL)

User ids are Supabase auth user ids (uuid). Every user-owned table has a userId column and a Row Level Security policy allowing access only where userId equals the signed-in user. Shared reference tables (Job, CompanyProfile) are read-only to signed-in users. IDs are cuid strings unless stated.

Enums: WorkMode (REMOTE, HYBRID, ONSITE) · JobType (FULL_TIME, PART_TIME, CONTRACT, INTERNSHIP) · ApplicationStatus (SAVED, APPLIED, INTERVIEWING, OFFER, REJECTED) · AlertFrequency (OFF, DAILY, WEEKLY) · AtsProvider (GREENHOUSE, LEVER, ASHBY) · ExperienceFit (UNDER, FIT, OVER).

Existing tables (built)

Table	Key columns
User	id (uuid, = auth user), email (unique), name, createdAt
Resume	id, userId, fileName, fileUrl (private storage path), rawText, parsedJson (JSON), isPrimary, createdAt

Table	Key columns
Preferences	userId (pk), targetRoles[], locations[], workModes[], minSalary, currency, experienceLevel, jobTypes[], keywordsInclude[], keywordsExclude[], noticePeriodDays, alertFrequency, alertEmail, alertWhatsapp, whatsappNumber, alertMinScore, lastAlertAt
Job	id, source, externalId, title, company, companySlug, companyLogoUrl, location, country, isRemote, employmentType, salaryMin, salaryMax, salaryCurrency (annual), descriptionHtml (sanitised), descriptionText, postedAt, applyUrl, publisher, attributionUrl, dedupeHash (unique), sourcesSeen[], skills[], isDirect, isSeed, fetchedAt. Unique (source, externalId).
Match	id, userId, jobId, score, stage1Score, aiScored, profileHash, reason, matchingSkills[], missingSkills[], experienceFit, notifiedAt. Unique (userId, jobId).
TailoredResume	id, userId, jobId, baseResumeId, contentJson, changes (JSON, null until tailored), skillGaps, pdfUrl, docxUrl, coverLetterText. Unique (userId, jobId, baseResumeId).
Application	id, userId, jobId, tailoredResumeId, status, position, appliedAt, notes, followUpDate. Unique (userId, jobId).
SavedSearch	id, userId, name, filters (JSON), alertFrequency, alertEmail, alertWhatsapp, lastAlertAt
CompanyProfile	id, slug (unique), name, website, logoUrl, description, industry, hqLocation, foundedYear, sizeRange, fundingStage, fundingNote, investors[], research links, dataSource, dataAsOf
CompanyReview	id, companyId, userId, rating (1-5), title, pros, cons, roleTitle, relationship. Unique (companyId, userId).
CompanyBoard	id, provider, slug, companyName, addedByUserId, enabled, lastFetchedAt, lastError, jobCount. Unique (provider, slug).

Table	Key columns
AiUsage	id, userId, feature, model, inputTokens, outputTokens, cacheReadTokens, createdAt (basis for per-user rate limits)
ReferralDraft	id, userId, jobId, contactName, contactRole, relationship, connectionNote, followUpMessage
AlertLog	id, userId, channel, status, jobCount, detail
FetchRun	id, source, startedAt, finishedAt, fetched, inserted, updated, error

New tables for outreach

Table	Key columns
ConnectedAccount	id, userId, provider (GMAIL, OUTLOOK), accountEmail, encryptedRefreshToken, scopes[], status (ACTIVE, REVOKED, ERROR), dailySendCap, warmupStage, connectedAt. Unique (userId, provider, accountEmail).
OutreachCampaign	id, userId, jobId (nullable for external links), sourceUrl, jobSnapshot (JSON), tailoredResumeId, companyId, mode (REVIEW_ALL, APPROVE_FIRST, AUTO), status (DRAFT, ACTIVE, PAUSED, COMPLETED, CANCELLED), createdAt, finishedAt
OutreachTarget	id, campaignId, fullName, title, roleType (HIRING_MANAGER, TEAM_LEAD, RECRUITER, WARM_PATH), companyDomain, email, emailVerification (VALID, RISKY, INVALID, UNKNOWN), linkedinUrl, dataProvider, sourceRef, confidence, state (DISCOVERED, VERIFIED, DRAFTED, APPROVED, SCHEDULED, SENT, REPLIED, BOUNCED, OPTED_OUT, COMPLETED, CANCELLED)

Table	Key columns
OutreachMessage	id, targetId, channel (EMAIL, LINKEDIN_NOTE, LINKEDIN_MESSAGE), step (1, 2, 3), subject, body, approvedAt, approvedTextHash, scheduledAt, sentAt, providerMessageId, threadId, state (DRAFT, APPROVED, SCHEDULED, SENT, FAILED, CANCELLED). Unique (targetId, channel, step) so a step is never sent twice.
LinkedInStep	id, targetId, state (NOT_STARTED, CONNECT_PENDING, ACCEPTED, MESSAGED, WITHDRAWN), requestedAt, acceptedAt, markedBy (USER or SYSTEM), expiresAt
Reply	id, messageId, receivedAt, snippet, classification (INTERESTED, REFERRAL, NOT_NOW, NO, OUT_OF_OFFICE, UNSUBSCRIBE, OTHER), suggestedResponse
SuppressionEntry	id, emailHash (unique), reason (OPT_OUT, BOUNCE, COMPLAINT, MANUAL), createdAt. Kept after account deletion.
AuditEvent	id, userId, action, entityType, entityId, detail (JSON), createdAt
CreditLedger	id, userId, delta, reason (PURCHASE, SPEND_TARGET, SPEND_VERIFY, REFUND, REPLY_BONUS), refType, refId, createdAt
ProviderUsage	id, userId, provider, operation, costMicros, createdAt

Indexes and constraints that matter: (userId, status) on Application and OutreachCampaign; (scheduledAt, state) on OutreachMessage for the sender worker; unique dedupeHash on Job; unique send key on OutreachMessage; emailHash unique on SuppressionEntry; all foreign keys cascade on user deletion except SuppressionEntry and AuditEvent hashes that must outlive the account.